

Consumer Behavior

**ECON 101: Introduction to
Microeconomics**

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Consumer Behavior

theory of consumer behavior Description of how consumers allocate incomes among different goods and services to maximize their well-being.

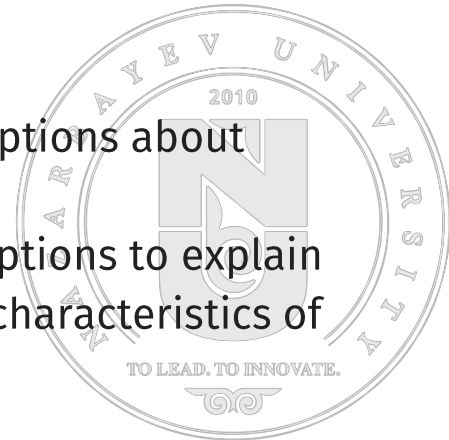
Consumer behavior is best understood in three distinct steps:

1. Consumer Preferences
2. Budget Constraints
3. Consumer Choices

WHAT DO CONSUMERS DO?

Recent models of consumer behavior incorporate more realistic assumptions about rationality and decision making.

A basic “workhorse” of economics, our model makes simplifying assumptions to explain much of what we actually observe regarding consumer choice and the characteristics of consumer demand.



Consumer Preferences

- **market basket (or bundle)** List with specific quantities of one or more goods.

MARKET BASKET	UNITS OF FOOD	UNITS OF CLOTHING
A	20	30
B	10	50
D	40	20
E	30	40
G	10	20
H	10	40

To explain the theory of consumer behavior, we will ask whether consumers *prefer* one market basket to another.

↳ Basic Object: Preference Relations



A model of consumer preferences: the Questionnaire "Q"

↳ List of questions of the following type:

$Q(x, y)$ — x, y are two distinct bundles

• How do you compare x to y ? Check one and only option

☐ I prefer x to y ($x \succ y$)

☐ I prefer y to x ($y \succ x$)

☐ I am indifferent ($x \sim y$)

A "model" of preferences is just a long list of questions of this type & their answers.

What is a legal answer to the questionnaire?

① For any question $Q(x, y)$ exactly one of the boxes is ticked.

We don't allow

- Refraining to answer
- Checking more than one box.

① is called completeness

② Suppose you have bundles x, y, z such that

$$Q(x, y) : x > y$$

$$Q(y, z) : y > z$$

$$Q(x, z) : x > z$$

$$Q(x, y) : x \sim y$$

$$Q(y, z) : y \sim z$$

$$Q(x, z) : x \sim z$$

② Transitivity

① + ② describe what we call "rational preferences"

③ Since we are talking about goods, which are assumed to be desirable, $\Rightarrow$ consumer always prefers more of a good i.e. the more the better.

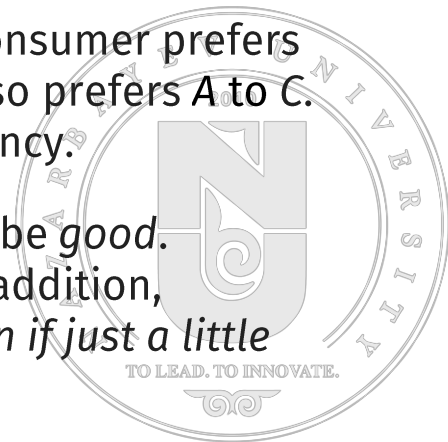
$$\left. \begin{array}{l} E = (30 \text{ food}, 40 \text{ clothing}) \\ A = (20 \text{ food}, 30 \text{ clothing}) \end{array} \right\} \begin{array}{l} Q(E, A) \\ \square \text{ I prefer } E \text{ to } A \end{array} \Rightarrow E \succ A$$

③ "more-is-better" OR MONOTONICITY

Consumer Preferences: Basic assumptions

Some Basic Assumptions about Preferences

- 1. Completeness:** Preferences are assumed to be *complete*. In other words, consumers can compare and rank all possible baskets. Thus, for any two market baskets *A* and *B*, a consumer will prefer *A* to *B*, will prefer *B* to *A*, or will be indifferent between the two. By *indifferent* we mean that a person will be equally satisfied with either basket.
- 2. Transitivity:** Preferences are *transitive*. Transitivity means that if a consumer prefers basket *A* to basket *B* and basket *B* to basket *C*, then the consumer also prefers *A* to *C*. Transitivity is normally regarded as necessary for consumer consistency.
- 3. More is better than less:** Goods are assumed to be desirable—i.e., to be *good*. Consequently, *consumers always prefer more of any good to less*. In addition, consumers are never satisfied or satiated; *more is always better, even if just a little better*.



Consumer Preferences: Indifference Curves

DESCRIBING INDIVIDUAL PREFERENCES

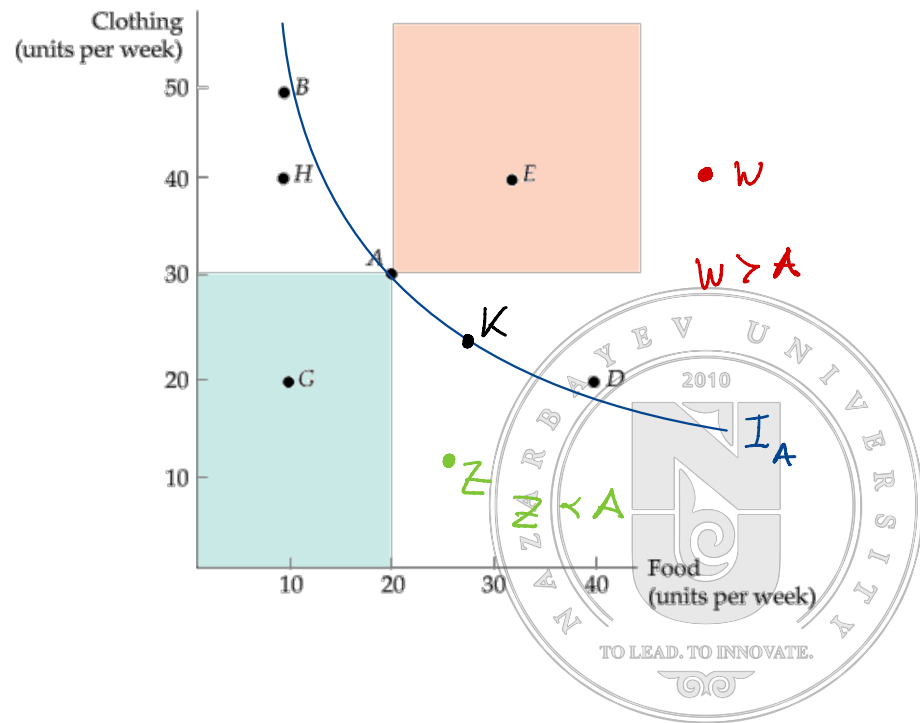
Because more of each good is preferred to less, we can compare market baskets in the shaded areas.

Basket A is clearly preferred to basket G, while E is clearly preferred to A.

However, A cannot be compared with B, D, or H without additional information.

$E \succ A \Rightarrow E \succ G$
 $A \succ G$ Transitivity

Indifference curve: For a bundle x , an indifference curve at x , is the set of all consumption bundles indifferent to x .



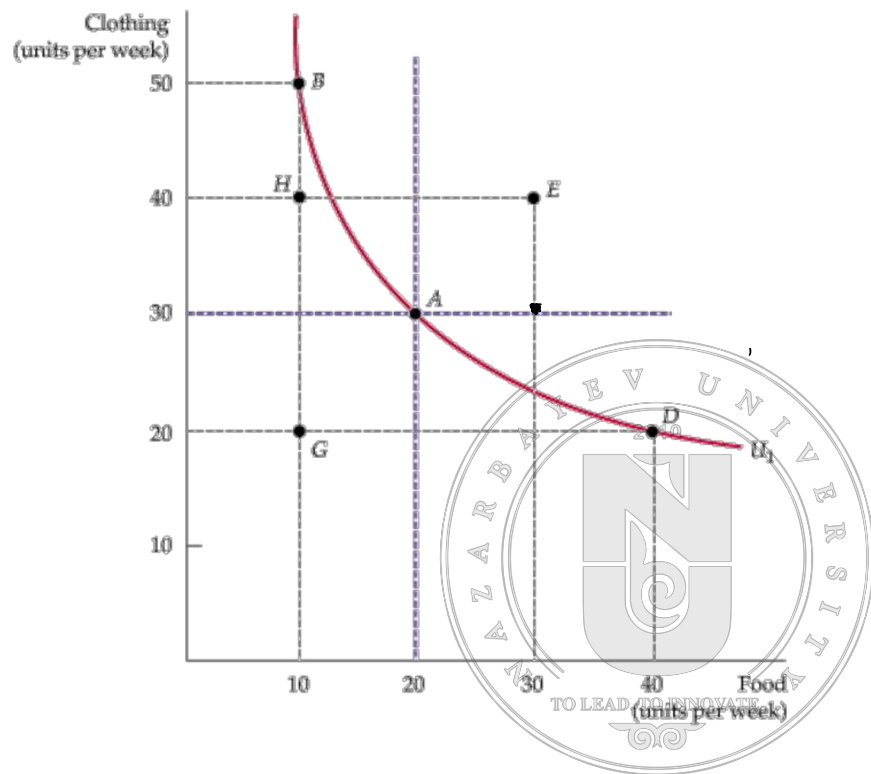
Consumer Preferences: Indifference Curves

Indifference Curves

Curve representing all combinations of market baskets that provide a consumer with the same level of satisfaction.

The indifference curve U_1 that passes through market basket A shows all baskets that give the consumer the same level of satisfaction as does market basket A; these include baskets B and D.

Our consumer prefers basket E, which lies above U_1 , to A, but prefers A to H or G, which lie below U_1 .

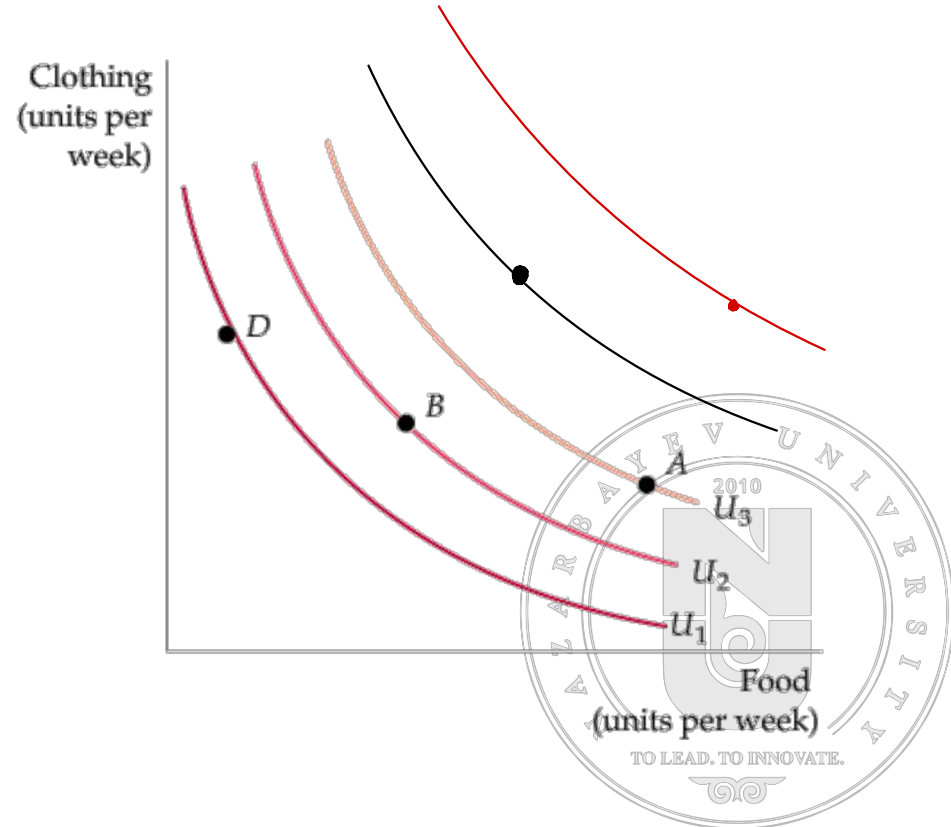


Consumer Preferences: Indifference Maps

Indifference Maps Graph containing a set of indifference curves showing the market baskets among which a consumer is indifferent.

An indifference map is a set of indifference curves that describes a person's preferences.

Any market basket on indifference curve U_3 , such as basket A , is preferred to any basket on curve U_2 (e.g., basket B), which in turn is preferred to any basket on U_1 , such as D .



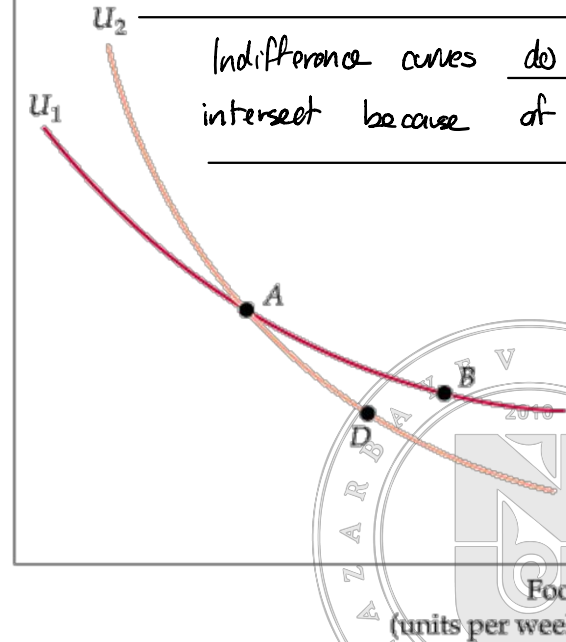
Consumer Preferences

INDIFFERENCE CURVES CANNOT INTERSECT

If indifference curves U_1 and U_2 intersect, one of the assumptions of consumer theory is violated.

According to this diagram, the consumer should be indifferent among market baskets A , B , and D . Yet B should be preferred to D because B has more of both goods.

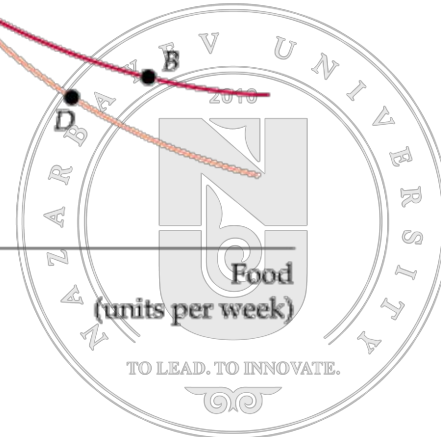
Clothing
(units per
week)



① $B \succ D$, $A \sim B$, $A \sim D \Rightarrow B \sim D$!

② $D \succ B$, $A \sim B$, $A \sim D \Rightarrow B \sim D$!

Indifference curves do not
intersect because of transitivity!

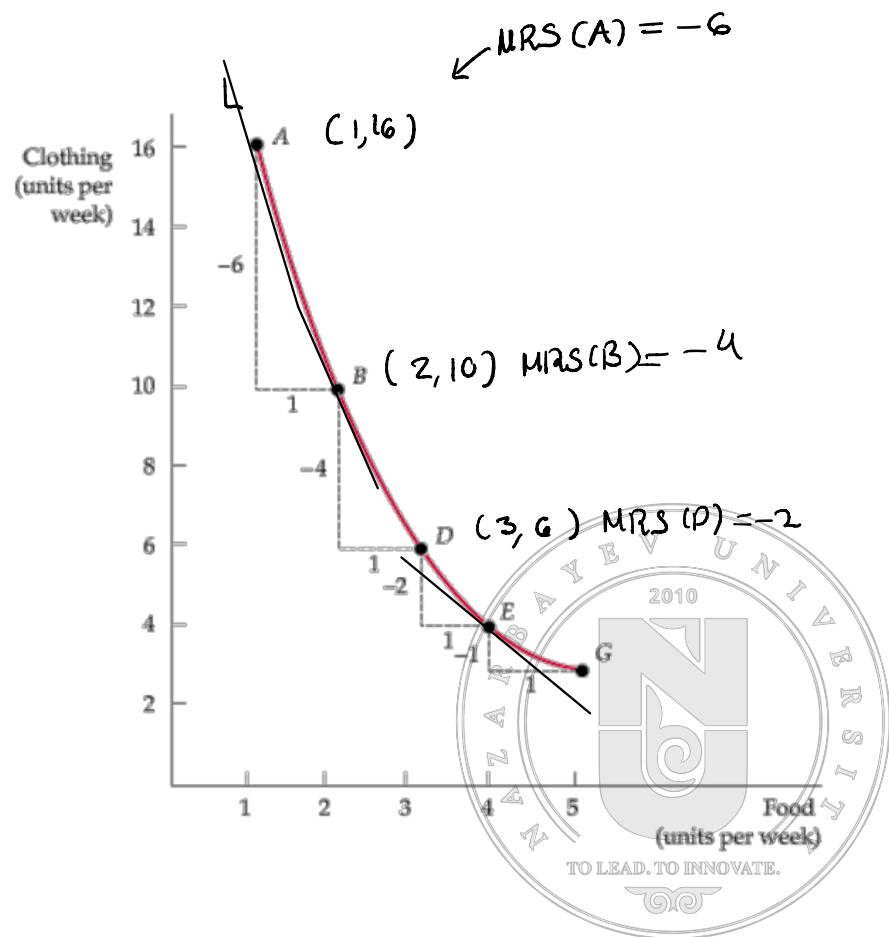


Shape of Indifference curves

THE MARGINAL RATE OF SUBSTITUTION

The magnitude of the slope of an indifference curve measures the consumer's marginal rate of substitution (MRS) between two goods.

In this figure, the MRS between clothing (C) and food (F) falls from 6 (between A and B) to 4 (between B and D) to 2 (between D and E) to 1 (between E and G).



The marginal rate of substitution

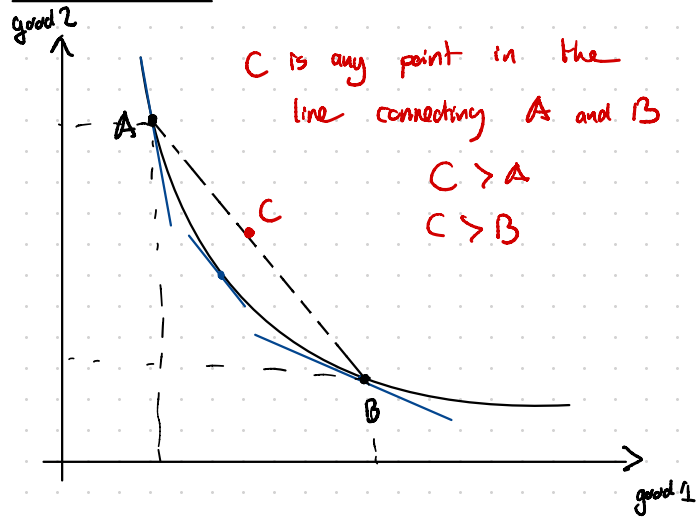
marginal rate of substitution (MRS) Maximum amount of a good that a consumer is willing to give up in order to obtain one additional unit of another good.

CONVEXITY

Observe that the MRS falls as we move down the indifference curve. The decline in the MRS reflects our fourth assumption regarding consumer preferences: a **diminishing marginal rate of substitution**. When the MRS diminishes along an indifference curve, the curve is convex.



Convexity : Preference for Balancedness

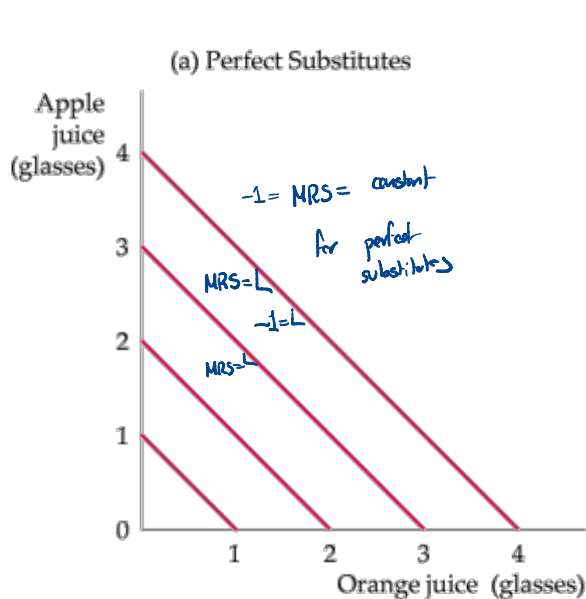


Consumer Preferences: some special cases

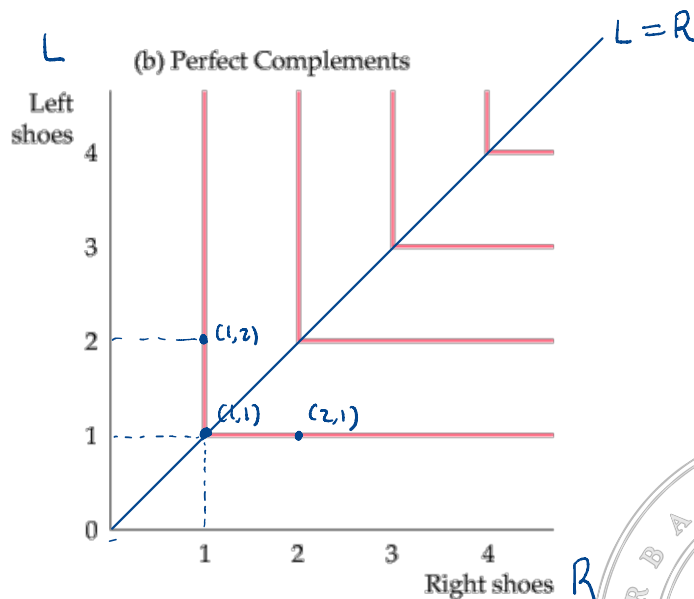
- **perfect substitutes** Two goods for which the marginal rate of substitution of one for the other is a constant.
- **perfect complements** Two goods for which the MRS is zero or infinite; the indifference curves are shaped as right angles.
- **bad** Good for which less is preferred rather than more.



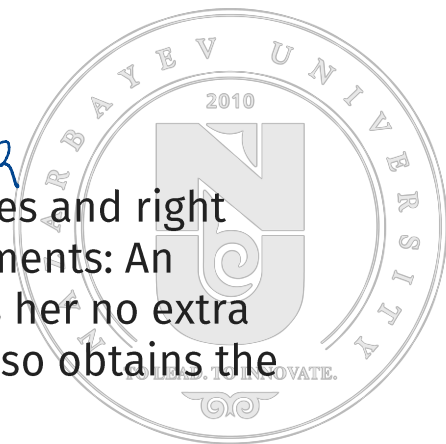
Perfect Substitutes and Perfect Complements



In **(a)**, Bob views orange juice and apple juice as perfect substitutes: He is always indifferent between a glass of one and a glass of the other.



In **(b)**, Jane views left shoes and right shoes as perfect complements: An additional left shoe gives her no extra satisfaction unless she also obtains the matching right shoe.

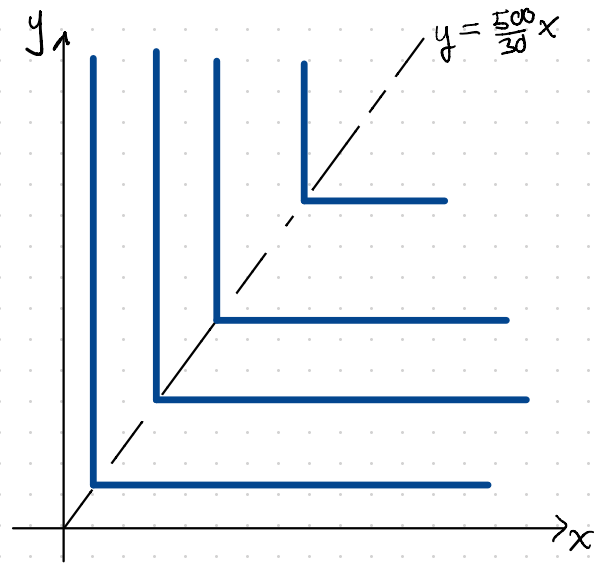


Perfect Complement → Example

Prof. Alejandro likes his coffee (pour-over) with the following recipe:

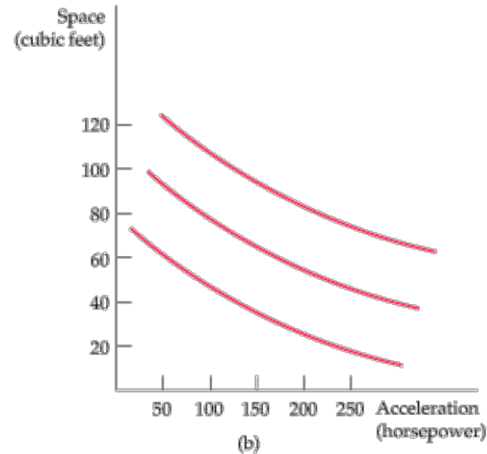
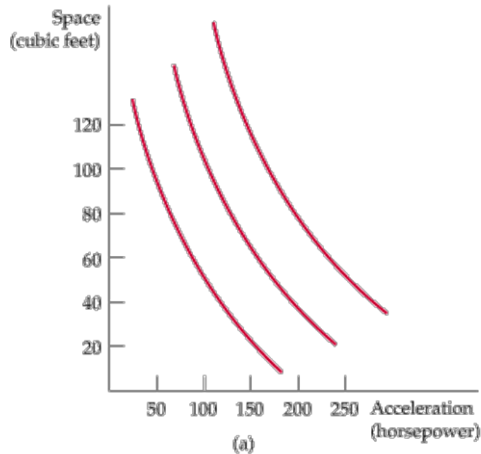
30 gr coffee : 500 gr. water
x y

$$y = 500 \left(\text{gr} \frac{x \text{ (gr)}}{30 \text{ gr}} \right) \quad / \quad y = \frac{500}{30} x$$



Example: designing new automobiles

- Preferences for automobile attributes can be described by indifference curves. Each curve shows the combination of acceleration and interior space that give the same satisfaction



Owners of Ford Mustang coupes **(a)** are willing to give up considerable interior space for additional acceleration.

The opposite is true for owners of Ford Explorers. They prefer interior space to acceleration **(b)**.



$$u: \mathbb{R}_+^2 \rightarrow \mathbb{R} : \text{ if } \begin{array}{l} x \succ y \Leftrightarrow u(x) > u(y) \\ y \succ x \Leftrightarrow u(y) > u(x) \\ x \sim y \Leftrightarrow u(x) = u(y) \end{array}$$

Utility and Utility functions

utility Numerical score representing the satisfaction that a consumer gets from a given market basket.

utility function Formula that assigns a level of utility to individual market baskets.

UTILITY FUNCTIONS AND INDIFFERENCE CURVES

A utility function can be represented by a set of indifference curves, each with a numerical indicator.

Example: This figure shows three indifference curves (with utility levels of 25, 50, and 100, respectively) associated with the utility function:

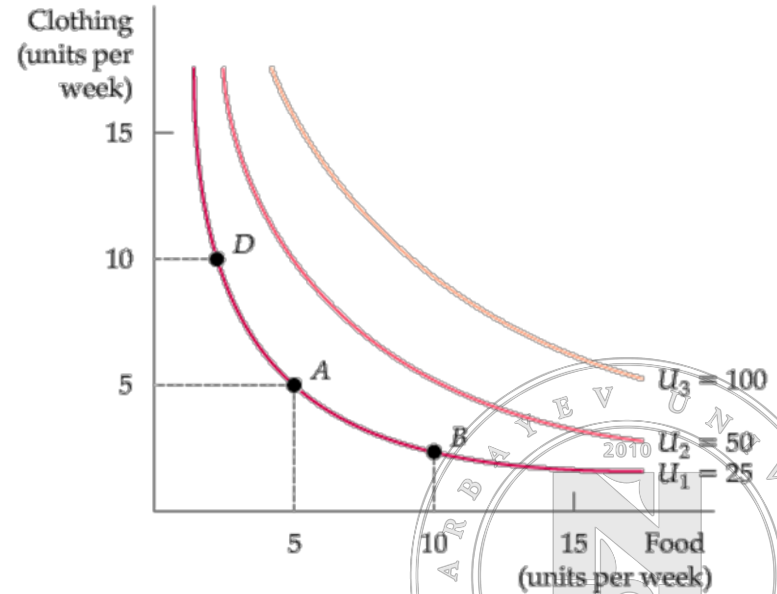
$$U(F, C) = FC$$

$$u(A) = u(5, 5) = 5 \cdot 5 = 25$$

$$I_c(A) \Rightarrow 25 = FC \Rightarrow C = \frac{25}{F}$$

$$I_c(x) = \left\{ \text{set of bundles } y \text{ such that } u(y) = u(x) \right\}$$

$$= \left\{ y \in \mathbb{R}_+^2 : u(y) = u(x) \right\}$$



Utility functions for perfect substitutes preferences

$$u(x_1, x_2) = ax_1 + bx_2, \quad a, b > 0$$

Ex: $x_1 = \text{coca}$ $x_2 = \text{pepsi}$

$$u(x_1, x_2) = 2x_1 + x_2$$

Fix a utility level $\bar{u}$

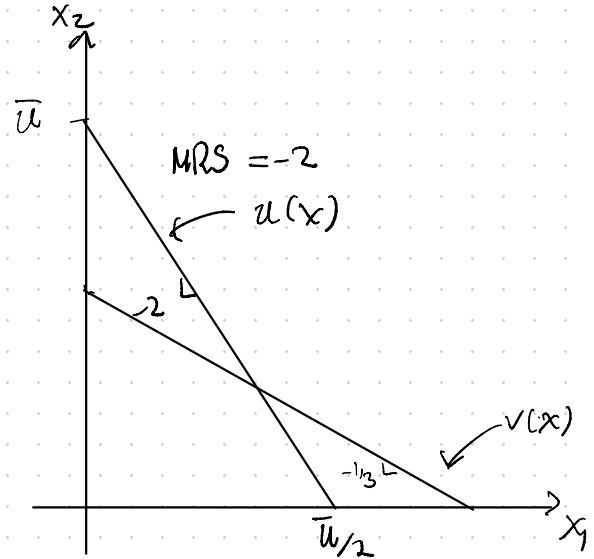
The IC: $\bar{u} = 2x_1 + x_2$

$$x_2 = \bar{u} - 2x_1$$

$$v(x_1, x_2) = x_1 + 3x_2$$

$$\bar{v} = x_1 + 3x_2$$

$$\frac{\bar{v} - x_1}{3} = x_2$$



Ordinal vs Cardinal Utility

ordinal utility function Utility function that generates a ranking of market baskets in order of most to least preferred.

cardinal utility function Utility function describing by how much one market basket is preferred to another.

$$\rightarrow x \succ y \Leftrightarrow u(x) > u(y) \Leftrightarrow \overset{f \circ u}{f(u(x)) > f(u(y))} \Leftrightarrow \overset{g \circ u}{g(u(x)) > g(u(y))}$$

$f: \mathbb{R} \rightarrow \mathbb{R}$ that increasing everywhere (monotonically increasing) } *monotonic transformation*
ex. $f(x) = x^3$
 $g(x) = \ln(x)$

$$u(F, C) = FC$$

$$V(F, C) = \ln(u(F, C)) = \ln(F) + \ln(C)$$



Budget Constraints

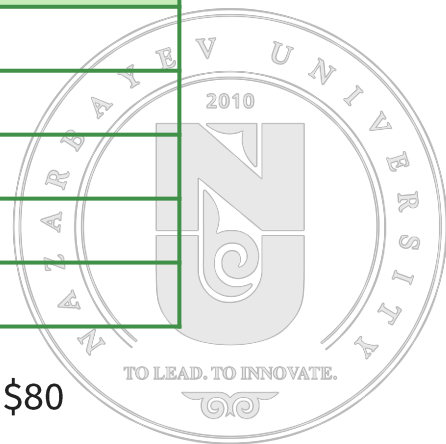
budget constraints Constraints that consumers face as a result of limited incomes.

- The Budget Line budget line All combinations of goods for which the total amount of money spent is equal to income.

$$P_F F + P_C C = I$$

MARKET BASKET	FOOD (F)	CLOTHING (C)	TOTAL SPENDING
A	0	40	\$80
B	20	30	\$80
D	40	20	\$80
E	60	10	\$80
G	80	0	\$80

Market baskets associated with the budget line $F + 2C = \$80$



Budget constraints

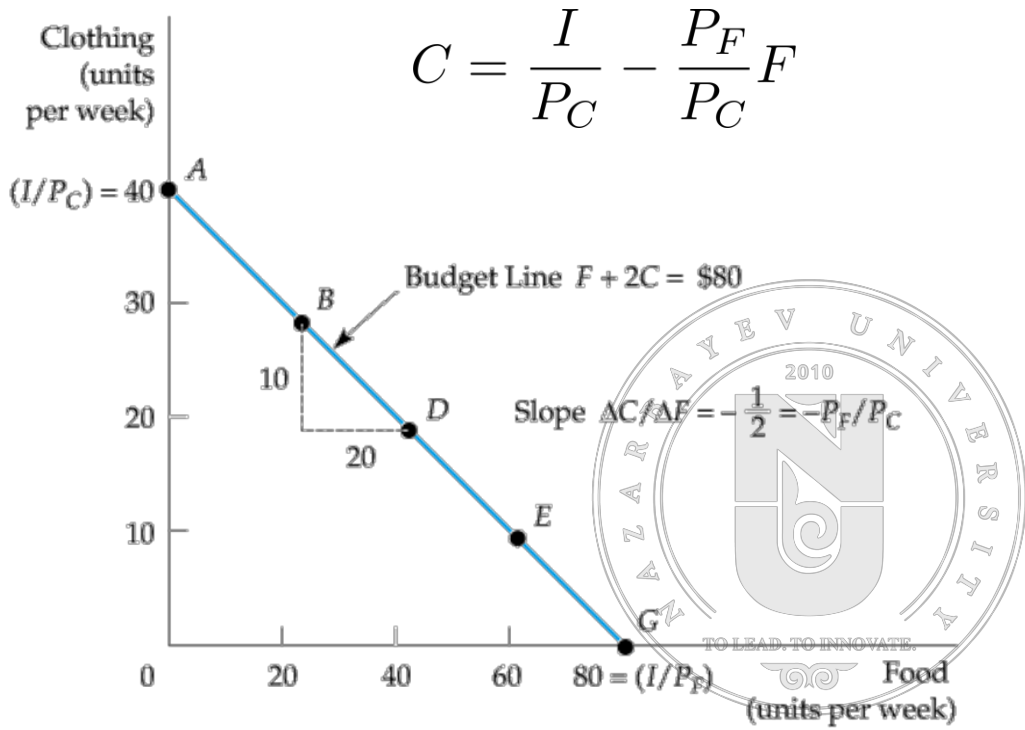
A BUDGET LINE

A budget line describes the combinations of goods that can be purchased given the consumer's income and the prices of the goods.

Line AG (which passes through points B, D, and E) shows the budget associated with an income of \$80, a price of food of $P_F = \$1$ per unit, and a price of clothing of $P_C = \$2$ per unit.

The slope of the budget line (measured between points B and D) is

$$-P_F/P_C = -10/20 = -1/2.$$



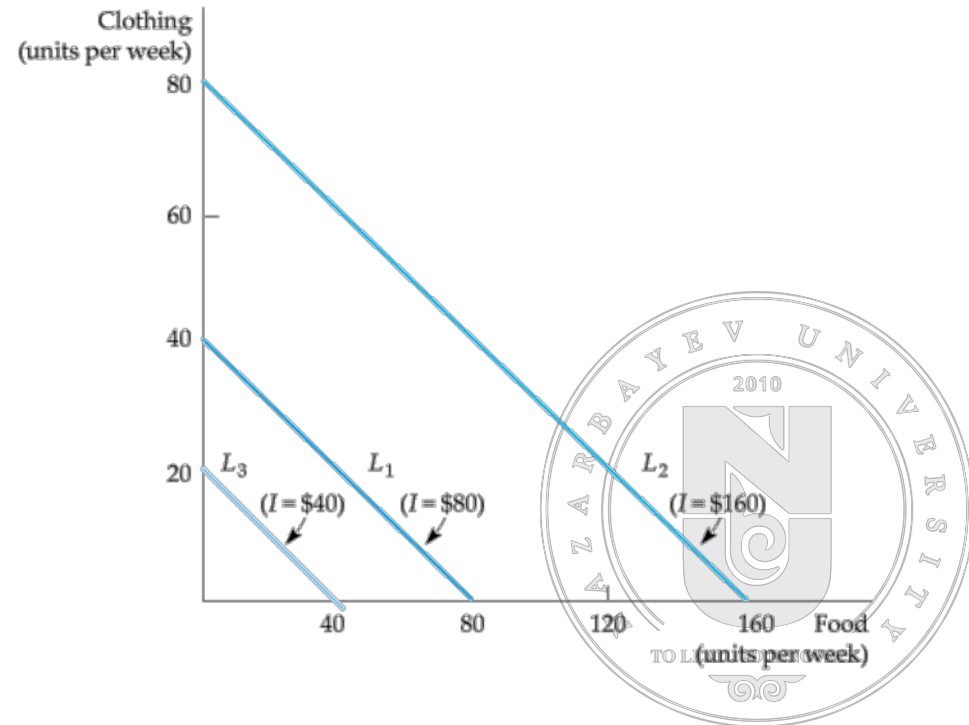
The Effects of changes in income and prices on the budget line

INCOME CHANGES

A change in income (with prices unchanged) causes the budget line to shift parallel to the original line (L_1).

When the income of \$80 (on L_1) is increased to \$160, the budget line shifts outward to L_2 .

If the income falls to \$40, the line shifts inward to L_3 .



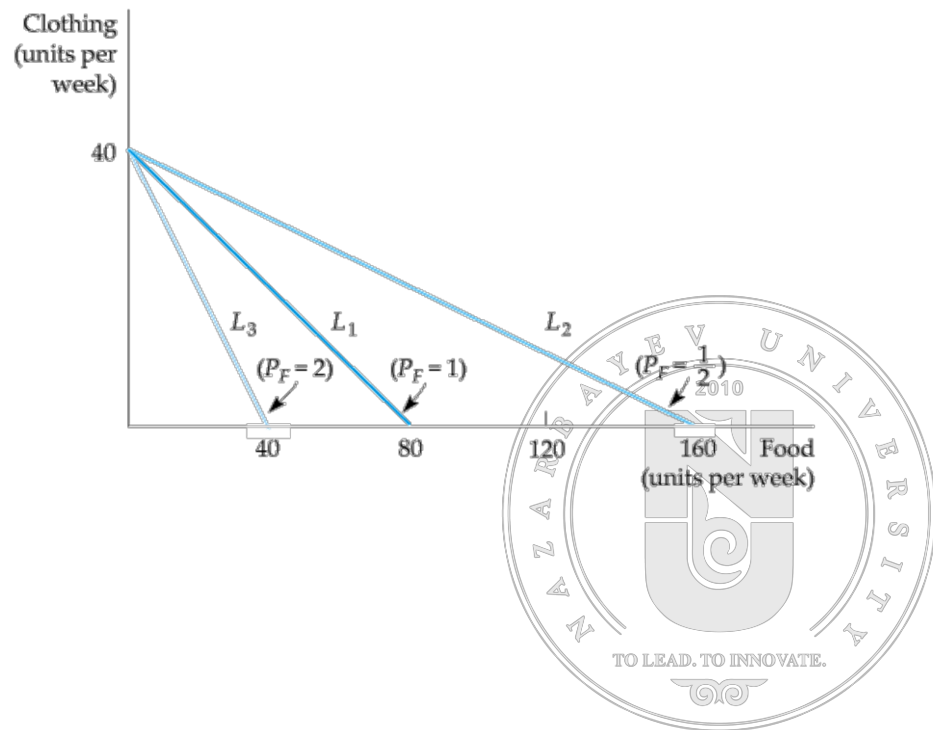
The Effects of changes in income and prices on the budget line

PRICE CHANGES

A change in the price of one good (with income unchanged) causes the budget line to rotate about one intercept.

When the price of food falls from \$1.00 to \$0.50, the budget line rotates outward from L_1 to L_2 .

However, when the price increases from \$1.00 to \$2.00, the line rotates inward from L_1 to L_3 .



Consumer Choice

Under the assumptions of:

1. Completeness
2. Transitivity
3. More is better

The maximizing market basket must satisfy two conditions:

1. It must be located on the budget line.
2. It must give the consumer the most preferred combination of goods and services.

If in addition, consumer preferences, as represented by the indifference curves, also have a strictly decreasing MRS (strict convexity), it must also be true that

$$\text{MRS} = \text{price ratio}$$



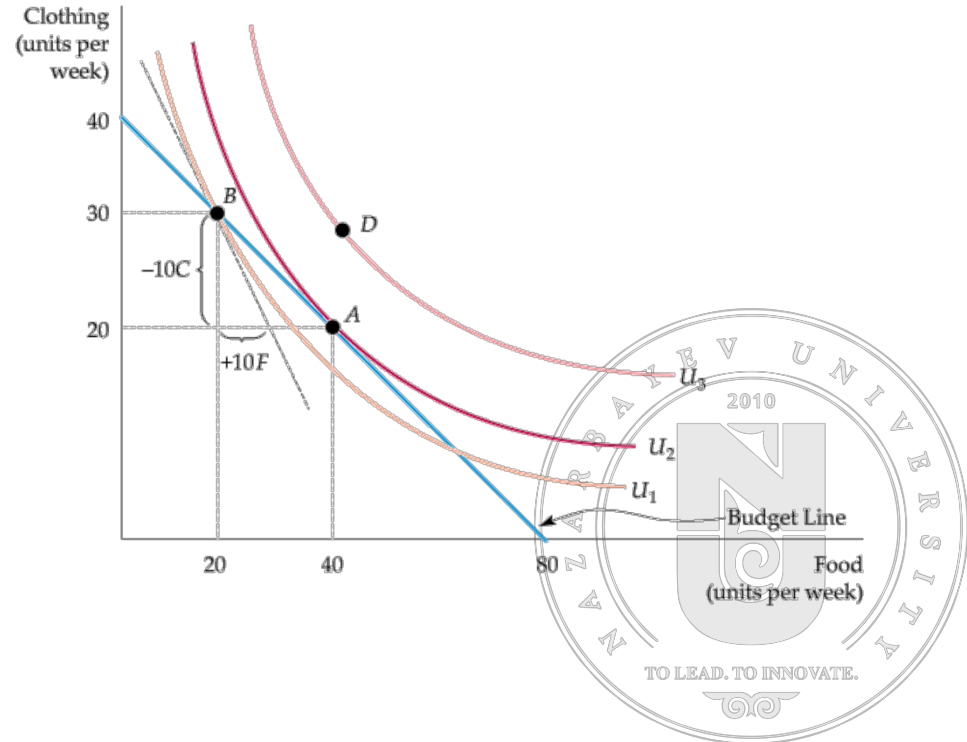
Consumer Choice

MAXIMIZING CONSUMER SATISFACTION

A consumer maximizes satisfaction by choosing market basket A. At this point, the budget line and indifference curve U_2 are tangent.

No higher level of satisfaction (e.g., market basket D) can be attained.

At A, the point of maximization, the MRS between the two goods equals the price ratio. At B, however, because the MRS [$-(-10/10) = 1$] is greater than the price ratio ($1/2$), satisfaction is not maximized.



Consumer Choice

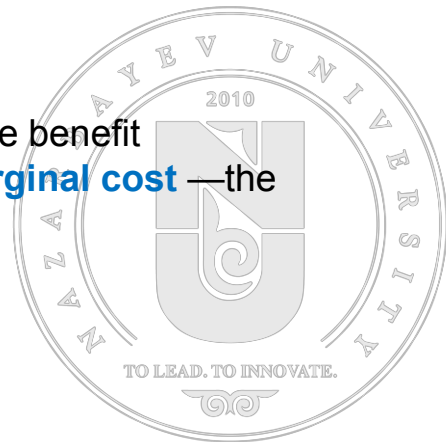
- Satisfaction is maximized (given the budget constraint) at the point where

$$MRS = \frac{P_F}{P_C}$$

marginal benefit Benefit from the consumption of one additional unit of a good.

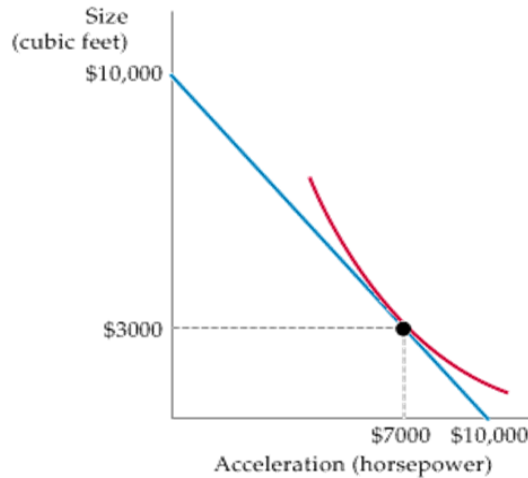
marginal cost Cost of one additional unit of a good.

So, we can then say that satisfaction is maximized when the **marginal benefit**—the benefit associated with the consumption of one additional unit of food—is equal to the **marginal cost**—the cost of the additional unit of food. The marginal benefit is measured by the MRS.



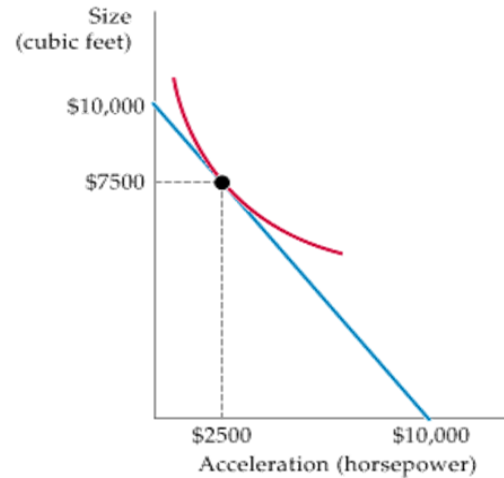
Example: Designing new automobiles

- Different preferences of consumer groups for automobiles can affect their purchasing decisions. Following up on the automobile example, we consider two groups of consumers planning to buy new cars.



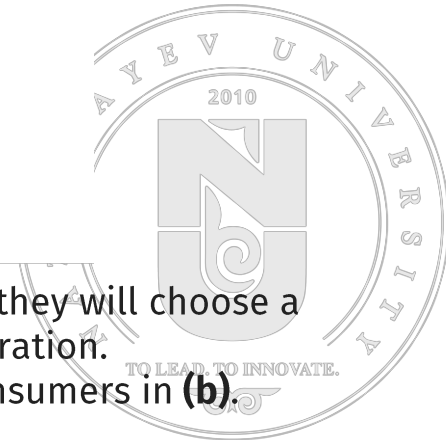
(a)

The consumers in **(a)** are willing to trade off a considerable amount of interior space for some additional acceleration.



(b)

Given a budget constraint, they will choose a car that emphasizes acceleration. The opposite is true for consumers in **(b)**.

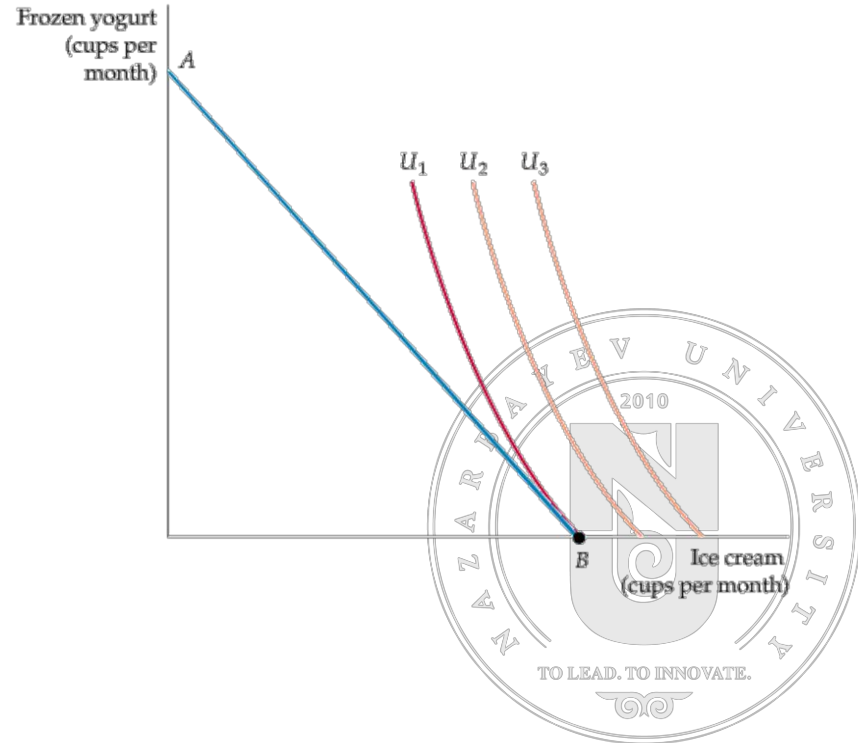


Special Cases: Corner solutions

corner solution Situation in which the marginal rate of substitution for one good in a chosen market basket is not equal to the slope of the budget line.

When a corner solution arises, the consumer maximizes satisfaction by consuming only one of the two goods.

Given budget line AB , the highest level of satisfaction is achieved at B on indifference curve U_1 , where the MRS (of ice cream for frozen yogurt) is greater than the ratio of the price of ice cream to the price of frozen yogurt.

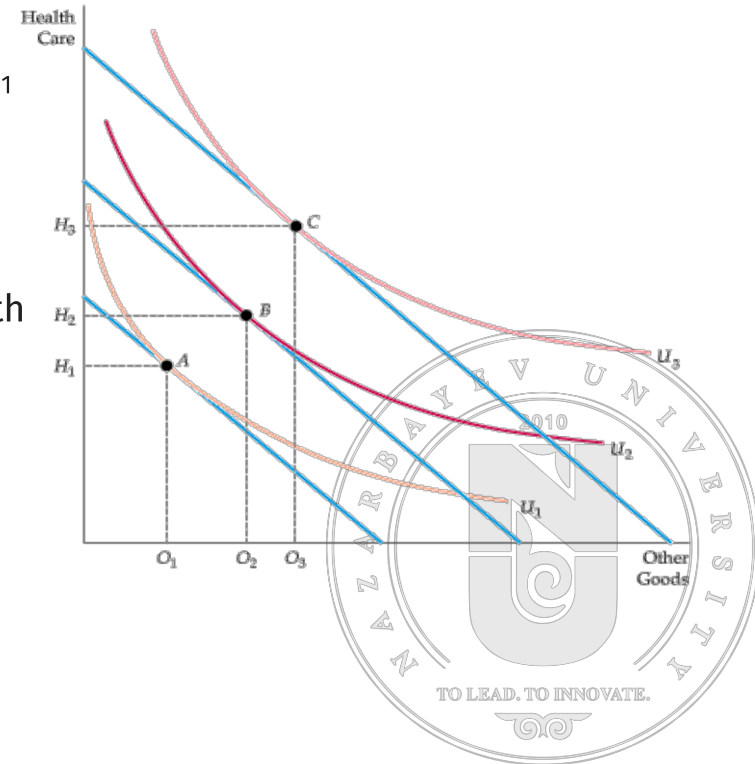


Example: Consumer choice of healthcare

These indifference curves show the trade-off between consumption of health care (H) versus other goods (O). Curve U_1 applies to a consumer with low income; given the consumer's budget constraint, satisfaction is maximized at point A.

As income increases the budget line shifts to the right, and curve U_2 becomes feasible. The consumer moves to point B, with greater consumption of both health care and other goods.

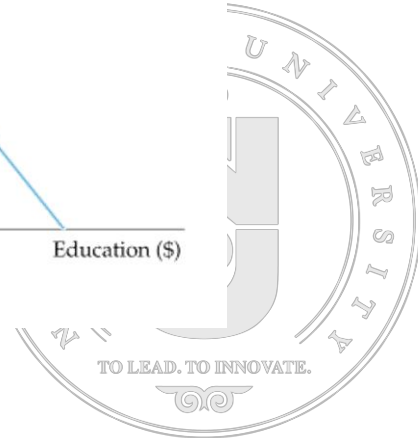
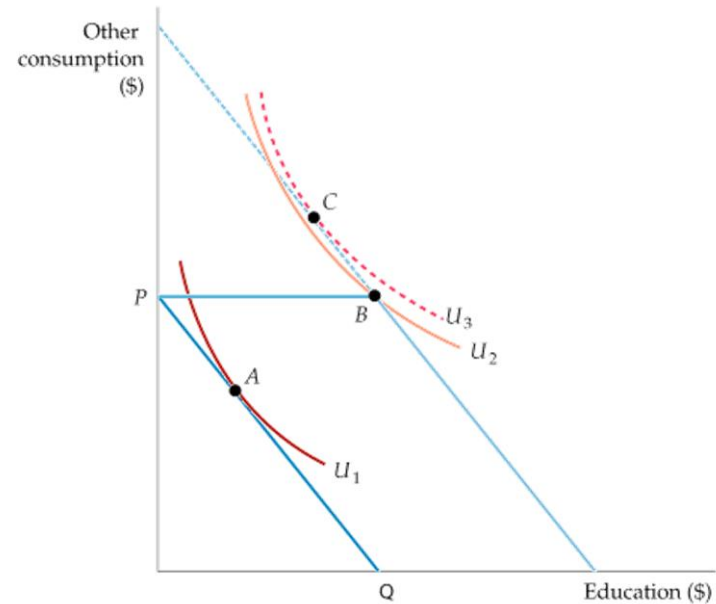
Curve U_3 applies to a high-income consumer, and implies less willingness to give up health care for other goods. Moving from point B to point C, the consumer's consumption of health care increases considerably (from H_2 to H_3), while her consumption of other goods increases only modestly (from O_2 to O_3).



Example: A college trust fund

When given a college trust fund that must be spent on education, the student moves from A to B , a corner solution.

If, however, the trust fund could be spent on other consumption as well as education, the student would be better off at C .



Revealed Preference

If a consumer chooses one market basket over another, and if the chosen market basket is more expensive than the alternative, then the consumer must prefer the chosen market basket.

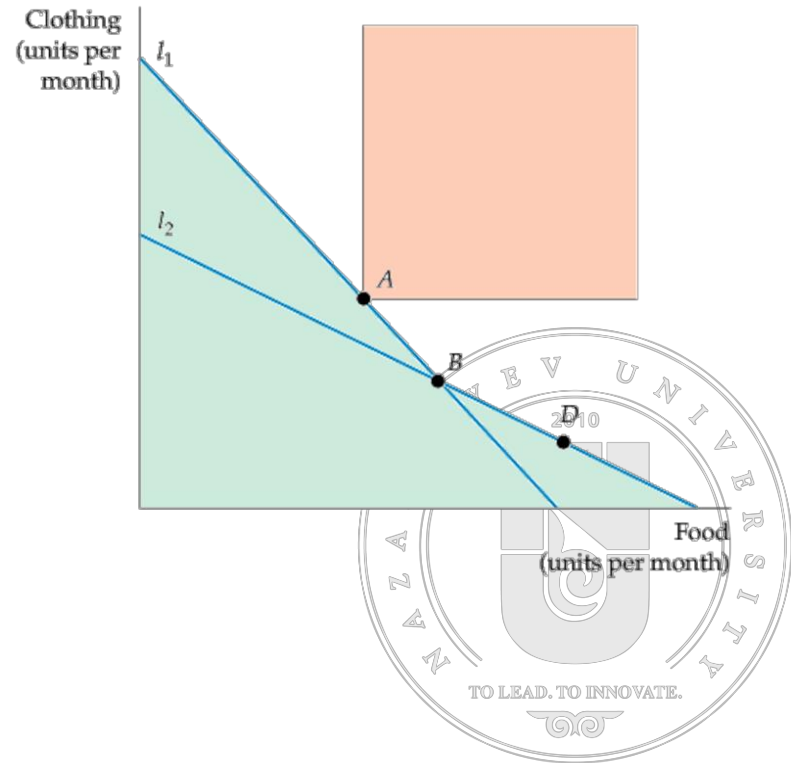
REVEALED PREFERENCE:

TWO BUDGET LINES

If an individual facing budget line l_1 chose market basket A rather than market basket B , A is revealed to be preferred to B .

Likewise, the individual facing budget line l_2 chooses market basket B , which is then revealed to be preferred to market basket D .

Whereas A is preferred to all market baskets in the green-shaded area, all baskets in the pink-shaded area are preferred to A .

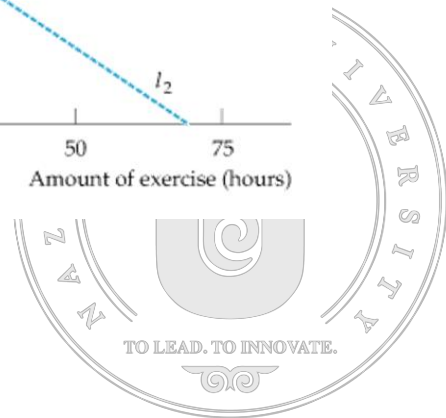
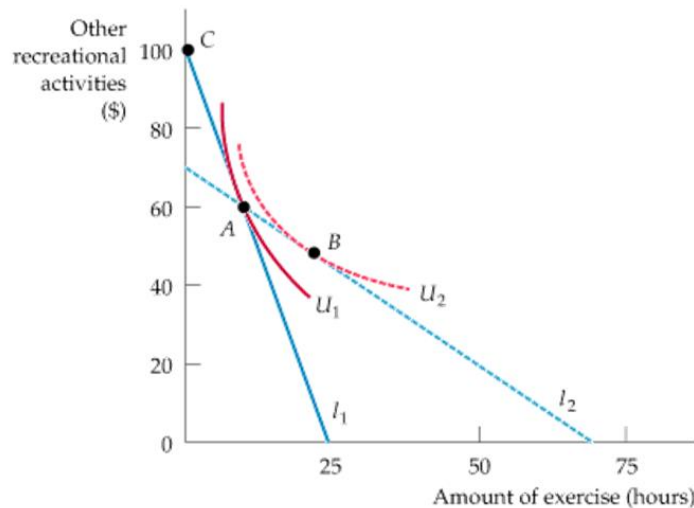


Example: Revealed Preference for Recreation

When facing budget line l_1 , an individual chooses to use a health club for 10 hours per week at point A.

When the fees are altered, she faces budget line l_2 .

She is then made better off because market basket A can still be purchased, as can market basket B, which lies on a higher indifference curve.



Marginal Utility and Consumer Choice

marginal utility (MU) Additional satisfaction obtained from consuming one additional unit of a good.

diminishing marginal utility Principle that as more of a good is consumed, the consumption of additional amounts will yield smaller additions to utility.

- At an indifference curve:

$$0 = MU_F (\Delta F) + MU_C (\Delta C)$$

$$-(\Delta C / \Delta F) = MU_F / MU_C$$

- Therefore:

$$MRS = MU_F / MU_C$$

$$MU_F / MU_C = P_F / P_C$$

$$MU_F / P_F = MU_C / P_C$$

- **equal marginal principle** Principle that utility is maximized when the consumer has equalized the marginal utility per dollar of expenditure across all goods.



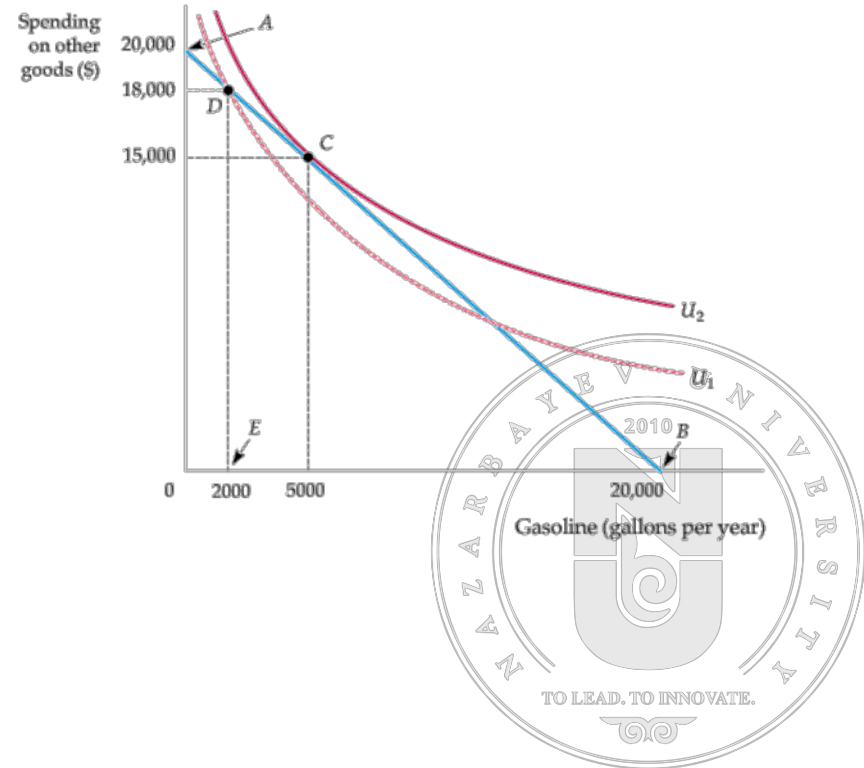
Marginal Utility and consumer choice

INEFFICIENCY OF GASOLINE RATIONING

When a good is rationed, less is available than consumers would like to buy. Consumers may be worse off.

Without gasoline rationing, up to 20,000 gallons of gasoline are available for consumption (at point *B*).

The consumer chooses point *C* on indifference curve U_2 , consuming 5000 gallons of gasoline. However, with a limit of 2000 gallons of gasoline under rationing, the consumer moves to *D* on the lower indifference curve U_1 .



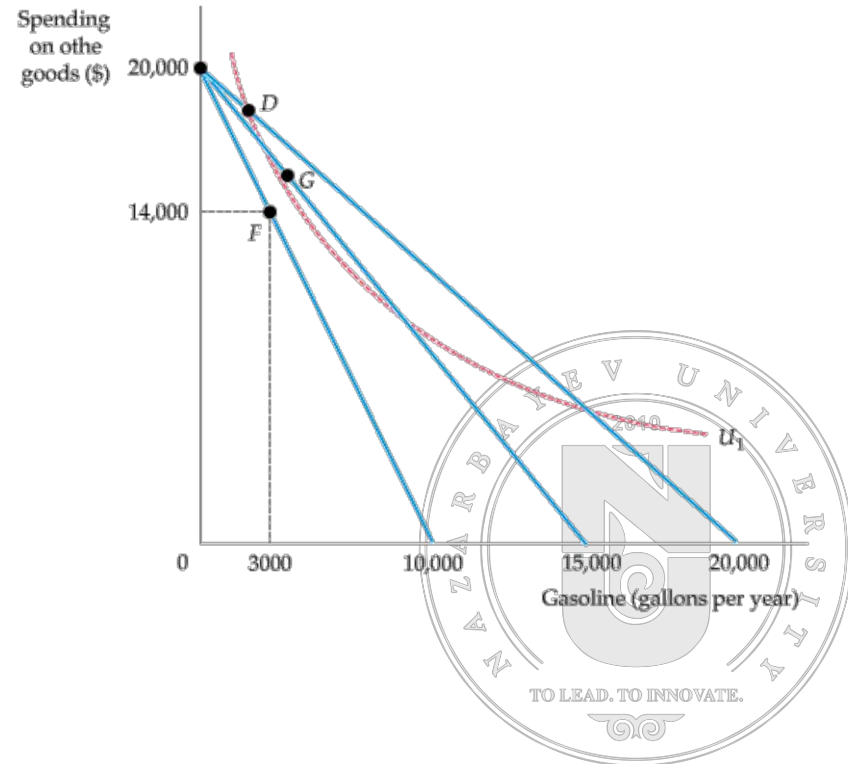
Marginal Utility and consumer choice

COMPARING GASOLINE RATIONING TO THE FREE MARKET

Some consumers will be worse off, but others may be better off with rationing. With rationing and a gasoline price of \$1.00, she buys the maximum allowable 2000 gallons per year, putting her on indifference curve U_1 .

Had the competitive market price been \$2.00 per gallon with no rationing, she would have chosen point F , which lies below indifference curve U_1 .

However, had the price of gasoline been only \$1.33 per gallon, she would have chosen point G , which lies above indifference curve U_1 .



Cost-of Living Indexes

- **cost-of-living index** Ratio of the present cost of a typical bundle of consumer goods and services compared with the cost during a base period.
- **ideal cost-of-living index** Cost of attaining a given level of utility at current prices relative to the cost of attaining the same utility at base-year prices.



Ideal Cost-of-Living Indexes

IDEAL COST-OF-LIVING INDEX

	2000	2010
Price of books	\$20/book	\$107.5/book
Number of books	15	9
Price of food	\$2.00/lb.	\$3.00/lb.
Pounds of food	100	215
Expenditure	\$500	\$1612.50

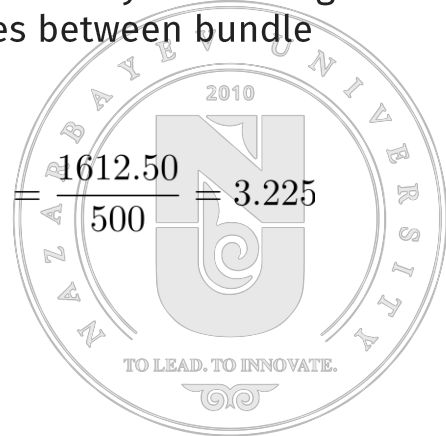
The year 2000 bundle is $A=(100,15)$. The 2010 bundle is $B=(215,9)$. At prices 2000 prices (2.00,20), bundle A costs \$500. At 2010 prices (3.00,107.50), bundle B costs \$1612.50.

Given utility function $U(F, B) = F^{0.4}B^{0.6}$, the utility from bundles A and B is the same.

Because of the price change between years 2000 and 2010, the consumer is substituting bundle A for bundle B, while holding her utility fixed, i.e., the satisfaction is obtained from bundles A and B.

The ideal price index is computed by calculating the ratio of total expenditures between bundle B and A, that is,

$$P_I = \frac{\text{Cost of } B \text{ in 2010 prices}}{\text{Cost of } A \text{ in 2000 prices}} = \frac{1612.50}{500} = 3.225$$



Cost-of-Living Indexes

IDEAL COST-OF-LIVING INDEX

	2000	2010
Price of books	\$20/book	\$107.5/book
Number of books	15	9
Price of food	\$2.00/lb.	\$3.00/lb.
Pounds of food	100	215
Expenditure	\$500	\$1612.50

Computing the ideal cost-of-living index requires a lot of knowledge regarding the consumer behavior, i.e., exactly how would the consumer react to a price change while holding her utility fixed. **This is usually not feasible.**

What can we do? Answer: Fix the market bundle at a particular year and compute its cost at base year prices and current year prices. **The index will be different depending in which year we fix the market bundle!**



Cost-of-Living Indexes

Laspeyres price index Amount of money at current year prices that an individual requires to purchase a bundle of goods and services chosen in a base year divided by the cost of purchasing the same bundle at base-year prices.

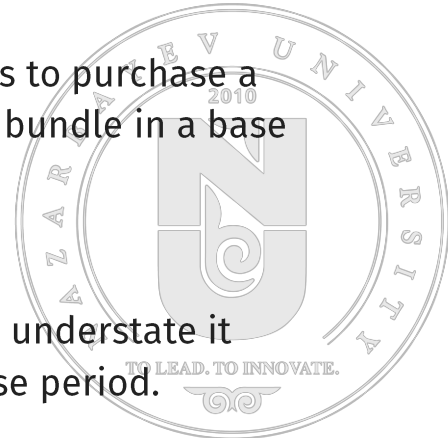
COMPARING IDEAL COST-OF-LIVING AND LASPEYRES INDEXES

The Laspeyres index overcompensates Rachel for the higher cost of living, and the Laspeyres cost-of-living index is, therefore, greater than the ideal cost-of-living index.

Paasche index Amount of money at current-year prices that an individual requires to purchase a current bundle of goods and services divided by the cost of purchasing the same bundle in a base year.

COMPARING THE LASPEYRES AND PAASCHE INDEXES

Just as the Laspeyres index will overstate the ideal cost of living, the Paasche will understate it because it assumes that the individual will buy the current year bundle in the base period.



Cost-of-Living Indexes

fixed-weight index Cost-of-living index in which the quantities of goods and services remain unchanged.

chain-weighted price index Cost-of-living index that accounts for changes in quantities of goods and services.





End of of this Lecture.